



HK IMO Training 2022-2023
Problem Set 2
For 1 October 2022



Problem 5. Let a, b, c, d be positive integers. Find the smallest possible value of

$$S = \left\lfloor \frac{a+b+c}{d} \right\rfloor + \left\lfloor \frac{b+c+d}{a} \right\rfloor + \left\lfloor \frac{c+d+a}{b} \right\rfloor + \left\lfloor \frac{d+a+b}{c} \right\rfloor.$$

Here, $\lfloor x \rfloor$ denotes the largest integer not exceeding x .

Problem 6. There are 99 people in a meeting. Some pairs of people shake hands with each other once. It is known that everyone has shaken hands for at least 81 times and at most 90 times. Prove that there are 10 people in the meeting who have shaken hands for the same number of times and there is another person who has shaken hands with all these 10 people.

Problem 7. In $\triangle ABC$, let M be the midpoint of BC . A variable point D lies inside $\triangle ABC$ such that $\angle BAD = \angle CDM$. Denote by Γ the circumcircle of $\triangle ABD$. Let $E \neq D$ be the intersection point of the line DM and Γ . Also, let F be the reflection of D in the tangent to Γ at B . Prove that the length of EF is independent of the position of D .

Problem 8. Solve the equation $2m^2 = n^{14} + 1$ in integers.